

Chapter 3: Photosynthesis

Photosynthesis is the process by which green plants convert light energy into chemical energy stored in glucose. It takes place in the chloroplast, an organelle that contains the green pigment chlorophyll.

Photosynthesis occurs in two stages. The first stage is the light-dependent reactions, which take place in the thylakoid membrane. Here, chlorophyll absorbs light and water is split in a step called photolysis, releasing oxygen as a by-product and producing ATP and NADPH.

The second stage is the Calvin cycle, also called the light-independent reactions, which takes place in the stroma. The Calvin cycle uses the ATP and NADPH from the first stage to fix carbon dioxide into glucose.

Chapter 3: Photosynthesis (continued)

The overall reaction can be summarised by the equation:
six carbon dioxide plus six water, in the presence of light, yields
one glucose plus six oxygen.

Three main factors affect the rate of photosynthesis: light intensity, carbon dioxide concentration, and temperature. When one of these is in short supply it becomes the limiting factor, capping the overall rate no matter how favourable the others are.